

Eucalyptus - Case Study Answers

1. Why would an organization choose Eucalyptus over other cloud platforms?

An organization would primarily choose Eucalyptus for its **native compatibility with Amazon Web Services (AWS) APIs**. This is a significant advantage for businesses that want to build a private cloud that mirrors the AWS environment. It allows for a seamless transition for developers already familiar with AWS tools and enables the creation of a hybrid cloud, combining on-premises infrastructure with the public AWS cloud for flexibility and data control.

2. How does AWS compatibility help in expanding cloud services?

AWS compatibility is crucial for expansion because it enables a **hybrid cloud strategy** with minimal friction. Organizations can develop and test applications in their private Eucalyptus cloud and then easily migrate or burst those workloads to the public AWS cloud to handle increased demand. This "build here, run there" approach allows companies to scale their services on-demand without being locked into a single vendor, leveraging the vast resources of AWS when needed.

3. What challenges might arise in managing a private cloud using Eucalyptus?

Managing a Eucalyptus private cloud can present several challenges:

- **Complexity:** The initial setup and ongoing maintenance can be complex, requiring skilled administrators with expertise in networking, storage, and virtualization.
- **Hardware Management:** The organization is responsible for procuring, managing, and upgrading all the underlying physical hardware, which can be costly and time-consuming.
- **Limited Community Support:** Compared to more widespread platforms like OpenStack, Eucalyptus has a smaller community, which can make troubleshooting and finding solutions more difficult.

Nimbus - Case Study Answers

1. What makes Nimbus more suitable for research and scientific applications?

Nimbus is particularly suitable for research and scientific applications because it was specifically designed to handle **compute-intensive and data-intensive scientific workloads**. It provides tools for managing virtual clusters and batch processing jobs, which are common in scientific research, simulations, and large-scale data analysis. Its focus on "Infrastructure-as-a-Service" for science gives researchers the control they need to configure complex computing environments.

2. How does resource sharing across institutions benefit cloud users?

Resource sharing across institutions creates a **federated cloud**, which is highly beneficial for users, especially in academia and research. It allows a researcher at one university to access specialized computing hardware (like GPUs or large memory nodes) at another institution. This pooling of resources leads to greater computational power, fosters collaboration on large-scale projects, and allows users to run experiments that would be impossible with their local resources alone.

3. What could be one limitation of Nimbus in large-scale deployments?

A key limitation of Nimbus in large-scale commercial deployments is its **niche focus**. It lacks the broad feature set, extensive ecosystem of third-party tools, and user-friendly

management interfaces found in more mainstream cloud platforms like OpenStack or OpenNebula. Its design is optimized for scientific batch jobs rather than diverse, enterprise-level web services, which can limit its flexibility in a corporate environment.

OpenNebula - Case Study Answers

1. Why is OpenNebula considered a good choice for startups and small businesses?

OpenNebula is considered a great choice for startups and small businesses because it is **lightweight, simple, and easy to deploy and manage**. Its low administrative overhead and modest hardware requirements reduce the total cost of ownership. This simplicity allows smaller organizations, which may not have large, specialized IT teams, to build a powerful and flexible private or hybrid cloud without the complexity of larger platforms.

2. How does hybrid cloud deployment help companies scale services?

Hybrid cloud deployment with OpenNebula allows companies to seamlessly **cloud-burst** to public cloud providers like AWS or Microsoft Azure. This means a company can run its normal workloads on its cost-effective private OpenNebula cloud. When faced with a sudden spike in demand, it can automatically provision additional resources from a public cloud to handle the load. This provides the best of both worlds: the cost control of a private cloud and the near-infinite scalability of the public cloud.

3. What key security measures should be applied in an OpenNebula-based environment?

Several key security measures are essential in an OpenNebula environment:

- **Network Isolation:** Use Virtual Networks (VNETs) and Access Control Lists (ACLs) to segment traffic and isolate virtual machines from each other, preventing unauthorized access.
- **User and Group Management:** Implement strong authentication and define granular permissions for users and groups to ensure they only have access to the resources they need (principle of least privilege).
- **Image Security:** Maintain a trusted catalog of virtual machine images that are hardened, regularly patched, and scanned for vulnerabilities before deployment.
- **Host Security:** The physical hypervisor nodes must be secured with firewalls, regular security updates, and intrusion detection systems to protect the foundation of the cloud.